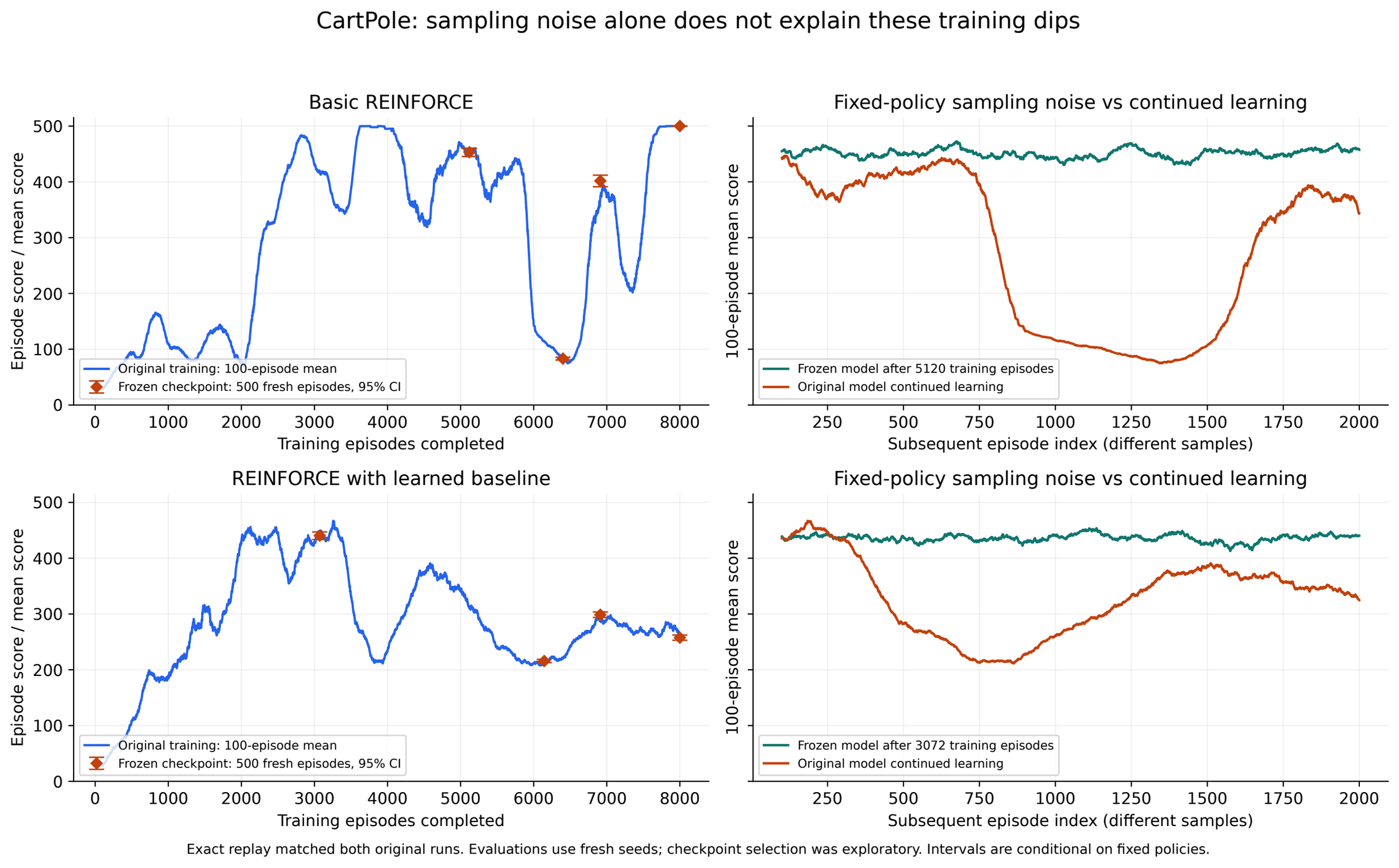

CartPole: sampling noise alone does not explain these training dips
Basic REINFORCE
Episode score / mean score
Training episodes completed
Original training: 100-episode mean
Frozen checkpoint: 500 fresh episodes, 95% CI
Fixed-policy sampling noise vs continued learning
100-episode mean score
Subsequent episode index (different samples)
Frozen model after 5120 training episodes
Original model continued learning
REINFORCE with learned baseline
Episode score / mean score
Training episodes completed
Original training: 100-episode mean
Frozen checkpoint: 500 fresh episodes, 95% CI
Fixed-policy sampling noise vs continued learning
100-episode mean score
Subsequent episode index (different samples)
Frozen model after 3072 training episodes
Original model continued learning
Exact replay matched both original runs. Evaluations use fresh seeds; checkpoint selection was exploratory. Intervals are conditional on fixed policies.